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SUPPLEMENTAL METHODS

Measurable/nonmeasurable disease status

Measurable disease status was based on modified Response Evaluation Criteria In Solid Tumors version 1.1 (RECIST) and Prostate Cancer Clinical Trials Working Group 3 (PCWG3) criteria. Up to 10 measurable target lesions were selected on the basis of their size and suitability for accurate repeated measurements, with a maximum of 5 per site. Computed tomography (CT) or magnetic resonance imaging scans were allowed. Per PCWG3, prostate and prostate bed lesions were considered nonmeasurable, as were soft-tissue lesions arising from bone.

Genomic screening

The following prespecified DNA damage repair (DDR) genes were evaluated during screening: *BRCA1*, *BRCA2*, *ATM*, *BARD1*, *BRIP1*, *CDK12*, *CHEK2*, *FANCA*, *NBN*, *PALB2*, *RAD51*, *RAD51B*, *RAD51C*, *RAD51D*, or *RAD54L*. A DDR gene alteration was determined through local or central genomic testing of plasma, archival tumor tissue, or screening tumor tissue. Once enrolled, patients with a known deleterious DDR gene alteration documented in the patient's medical record were also asked to submit archival tumor tissue, if available.

The current analysis included patients with a deleterious *BRCA1* or *BRCA2* (BRCA) alteration identified prior to enrollment, regardless of presence of a co-occurring alteration of a non-BRCA DDR gene, and who received ≥ 1 dose of rucaparib 600 mg.

Central germline testing was performed by Color Genomics. Patients with a BRCA alteration detected by the germline test were categorized as 'germline'; patients with a BRCA alteration that was not detected by the germline test were categorized as 'somatic'.

Alteration zygosity was determined by copy-number analysis of tissue next-generation sequencing data. Biallelic alterations included homozygous copy-number loss reported by local (n=6) or central (n=15) testing and truncating alterations with loss of heterozygosity (LOH) of the other allele determined from central testing (n=15). Monoallelic alterations (n=9) included truncating alterations with no LOH of the other allele. Loss of the second allele through other mechanisms could not be ruled out.

Safety assessments

Serious adverse events (SAEs) related to a prescreening or screening procedure that occurred after signing of an informed consent form were recorded. TEAEs were monitored from the first dose of rucaparib until 28 days after the last dose. Ongoing SAEs, adverse events of special interest, and grade 3 or 4 adverse events were followed until resolution or stabilization. After the 28-day window, only treatment-related SAEs, and all adverse events of special interest irrespective of causality were required to be reported.

SUPPLEMENTAL RESULTS

Deaths

There were 3 deaths due to treatment-emergent adverse events, including 1 each due to pneumonia and prolonged QT, both considered unrelated to rucaparib, and 1 due to acute respiratory distress syndrome, considered related to rucaparib by the investigator. Brief case histories for each are provided below.

Pneumonia: Patient died due to pneumonia 4 months after initiating rucaparib, after 1 dose reduction due to fatigue. The patient was seen in the clinic for his scheduled study visit and reported increased weakness and worsening cough. A CT scan of the chest, abdomen, and pelvis on the same day showed new dependent consolidation in the right upper lobe and left lower lobe, which was indicative of infection or aspiration. Treatment with antibiotics was given, but he died 5 days later. Medical history included hypothyroidism, hypertension, fatigue, weakness, and cough. Concomitant medications included levothyroxine, atenolol, amlodipine, calcium-carbonate with vitamin D, fosinopril, and dexamethasone.

Prolonged QT: Patient was hospitalized due to an electrocardiogram showing prolonged QT and runs of ventricular tachycardia (torsades de pointes), which occurred 267 days after initiating treatment with rucaparib. On day 6 of admission, the patient died. The treating cardiologist reported the primary cause of death as metastatic prostate cancer with “antecedent causes” reported as renal impairment and ischemic heart disease. Long QT syndrome of 5 years duration was listed as an “other significant condition” contributing to the patient’s death. In this case, the QTc interval at the time of the event (486 ms), occurring 9 months after starting rucaparib and after 2 dose reductions due to scalp dermatitis and symptomatic anemia, had not increased from baseline (509 ms). The patient had multiple risk factors for QT prolongation and torsades de pointes including a history of prolonged QT for >5 years; ischemic heart disease and recent myocardial infarction; and concomitant use of medications included metoclopramide, verapamil, and furosemide which are also associated with QT interval prolongation and torsades de pointes.¹⁻³

Acute respiratory distress syndrome: Patient died due to acute respiratory distress syndrome 129 days after initiating rucaparib. The patient had a history of dyspnea and pulmonary fibrosis of mixed origin, including exposure to asbestos and ongoing treatment with amiodarone, and had atypical pneumopathy on baseline CT scan. The investigator also considered infectious pneumopathy, disease progression (carcinomatous lymphangitis), and amiodarone use as other potential causes of the fatal respiratory failure.

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SUPPLEMENTAL FIGURES & TABLES

Supplementary Table S1. Baseline measurable disease status per investigator in TRITON2 patients with a BRCA alteration.

	N=115
Measurable disease (per investigator), n (%) ^a	
Measurable disease	65 (56.5)
Only measurable nodal disease	36 (55.4)
Measurable visceral ± nodal disease	29 (44.6)
Nonmeasurable disease	50 (43.5)
Bone-only disease	38 (76.0)
Other non-measurable disease	12 (24.0)
Visit cutoff date: December 23, 2019. ^a Based on modified RECIST/PCWG3 criteria. PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.	

Supplementary Table S2. Baseline patient and disease characteristics in TRITON2 patients with a BRCA1 or BRCA2 alteration.

	BRCA1 (n=13)	BRCA2 (n=102)
Age, median (range), years	73.0 (53–82)	72.0 (50–88)
Race, n (%)		
White	10 (76.9)	75 (73.5)
Black or African American	1 (7.7)	6 (5.9)
Other	0	2 (2.0)
Unknown	2 (15.4)	19 (18.6)
ECOG PS, n (%)		
0	6 (46.2)	31 (30.4)
1	6 (46.2)	70 (68.6)
≥2	1 (7.7)	1 (1.0)
Baseline PSA, median (range), ng/mL	91.0 (7.1–4669.0)	52.1 (0–4782.0)
Baseline ALP, median (range), U/L	95.5 (42.0–664.0)	98.0 (34.0–5490.0)
Baseline albumin, median (range), g/L	40.0 (29.0–43.0)	39.0 (24.0–49.0)
Baseline LDH, median (range), U/L	238 (144–864)	242 (84–2058)
Gleason score ≥8 at diagnosis, n (%)	7 (53.8)	70 (68.6)
No. prior CRPC therapies, median (range) ^a	3 (2–3)	2 (1–8)
Prior therapies, n (%) ^b		
Next-generation AR-directed therapy		
Abiraterone acetate	9 (69.2)	65 (63.7)
Enzalutamide	9 (69.2)	73 (71.6)
Apalutamide	0	3 (2.9)
Darolutamide	0	0
≥2 AR-directed therapies	5 (38.5)	37 (36.3)
Docetaxel (castration-sensitive)	0	17 (16.7)
Docetaxel (castration-resistant)	13 (100)	95 (93.1)
Cabazitaxel	0	8 (7.8)

Sipuleucel-T	0	12 (11.8)
Radium-223	1 (7.7)	13 (12.7)
Measurable disease (per blinded IRR), n (%) ^c		
Measurable disease	9 (69.2)	53 (52.0)
Only measurable nodal disease	6 (66.7)	35 (66.0)
Measurable visceral ± nodal disease	3 (33.3)	18 (34.0)
Nonmeasurable disease	4 (30.8)	49 (48.0)
Bone-only disease	3 (75.0)	33 (67.3)
Other non-measurable disease	1 (25.0)	16 (32.7)
Site of metastases (per blinded IRR), n (%)		
Hepatic	2 (15.4)	12 (11.8)
Lung	1 (7.7)	13 (12.7)
≥10 bone lesions (per blinded IRR), n (%)	6 (46.2)	48 (47.1)
Visit cutoff date: December 23, 2019. ^a Does not include luteinizing hormone–releasing hormone analogues, first-generation antiandrogens, hormones, corticosteroids, bone-targeted agents, hematopoietic growth factors, or docetaxel given for castration-sensitive disease (per protocol, docetaxel in this setting was not counted towards the number of prior mCRPC therapies for study eligibility). ^b Categories are not mutually exclusive. ^c Based on modified RECIST/PCWG3 criteria. ALP, alkaline phosphatase; AR, androgen receptor; CRPC, castration-resistant prostate cancer; ECOG PS, Eastern Cooperative Oncology Group Performance Status; LDH, lactate dehydrogenase; IRR, independent radiology review; PCWG3, Prostate Cancer Clinical Trials Working Group 3; PSA, prostate-specific antigen; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.		

Supplementary Table S3. Genomic characteristics of patients.

	BRCA1 (n=13)	BRCA2 (n=102)	All BRCA (n=115)
Germline/somatic status, n (%)			
Germline	5 (38.5)	39 (38.2)	44 (38.3)
Somatic	8 (61.5)	63 (61.8)	71 (61.7)
Alteration zygosity, n (%) ^a			
Biallelic	2 (15.4)	34 (33.3)	36 (31.3)
Monoallelic	3 (23.1)	6 (5.9)	9 (7.8)
Unknown zygosity	8 (61.5)	62 (60.8)	70 (60.9)
Visit cutoff date: December 23, 2019. ^a Due to the limited number of tissue samples received (66/115 [57.4%] patients), alteration zygosity could only be determined in 45 (39.1%) patients based on the availability of sufficiently high-quality next generation sequencing data from tissue for copy-number analysis.			

Supplementary Table S4. BRCA alteration types observed.

	BRCA1 (n=13)	BRCA2 (n=102)	All BRCA (n=115)
Alteration type, n (%)			
Frameshift mutation	5 (38.5)	51 (50.0)	56 (48.7)
Homozygous loss	1 (7.7)	20 (19.6)	21 (18.3)
Missense mutation	3 (23.1)	8 (7.8)	11 (9.6)
Nonsense mutation	0	14 (13.7)	14 (12.2)
Rearrangement	3 (23.1)	7 (6.9)	10 (8.7)
Splice site mutation	1 (7.7)	2 (2.0)	3 (2.6)
Visit cutoff date: December 23, 2019.			

Supplementary Table S5. Confirmed ORR by BRCA1 or BRCA2 alteration in the IRR-evaluable population.

	BRCA1 (n=9)	BRCA2 (n=53)
Confirmed ORR, n (%) [95% CI] ^a	3 (33.3) [7.5-70.1]	24 (45.3) [31.6-59.6]
Complete response, n (%)	1 (11.1)	6 (11.3)
Partial response, n (%)	2 (22.2)	18 (34.0)
Stable disease, n (%)	4 (44.4)	24 (45.3)
Progressive disease, n (%)	2 (22.2)	4 (7.5)
Not evaluable, n (%)	0	1 (1.9)
Visit cutoff date: December 23, 2019. ^a Per modified RECIST/PWCG3 criteria. IRR, independent radiology review; ORR, objective response rate; PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.		

Supplementary Table S6. Confirmed ORR by germline or somatic status of alteration in the IRR-evaluable population.

	Germline (n=21)	Somatic (n=41)
Confirmed ORR, n (%) [95% CI] ^a	9 (42.9) [21.8-66.0]	18 (43.9) [28.5-60.3]
Complete response, n (%)	0	7 (17.1)
Partial response, n (%)	9 (42.9)	11 (26.8)
Stable disease, n (%)	11 (52.4)	17 (41.5)
Progressive disease, n (%)	1 (4.8)	5 (12.2)
Not evaluable, n (%)	0	1 (2.4)
Visit cutoff date: December 23, 2019. ^a Per modified RECIST/PWCG3 criteria. IRR, independent radiology review; ORR, objective response rate; PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.		

Supplementary Table S7. Confirmed ORR by alteration zygosity in the IRR-evaluable population.

	Biallelic (n=23)	Monoallelic (n=4)	Unknown zygosity (n=35)
Confirmed ORR, n (%) [95% CI] ^a	12 (52.2) [30.6-73.2]	2 (50.0) [6.8-93.2]	13 (37.1) [21.5-55.1]
Complete response, n (%)	5 (21.7)	0	2 (5.7)
Partial response, n (%)	7 (30.4)	2 (50.0)	11 (31.4)
Stable disease, n (%)	10 (43.5)	1 (25.0)	17 (48.6)
Progressive disease, n (%)	1 (4.3)	1 (25.0)	4 (11.4)
Not evaluable, n (%)	0	0	1 (2.9)
Visit cutoff date: December 23, 2019. ^a Per modified RECIST/PWCG3 criteria. ORR, objective response rate; PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.			

Supplementary Table S8. Reasons for censoring in the analysis of radiographic progression-free survival by blinded IRR assessment.

Reason for censoring	N=115
Ongoing on study	28
Discontinued for the following reasons	
Disease progression by investigator without radiographic disease progression by IRR	22
Withdrawal of consent	5
Adverse event	4
Physician decision	2
Other	1
Visit cutoff date: December 23, 2019. IRR, independent radiology review.	

Supplementary Table S9. Safety summary.

Summary of TEAEs, n (%)	N=115
≥1 TEAE	114 (99.1)
≥1 grade 3 or higher TEAE	70 (60.9)
TEAE leading to dose modification	73 (63.5)
TEAE leading to treatment interruption ^a	65 (56.5)
TEAE leading to dose reduction ^b	47 (40.9)
TEAE leading to discontinuation ^c	9 (7.8)
TEAE leading to death	3 (2.6)
Visit cutoff date: September 13, 2019. TEAEs were graded according to NCI CTCAE version 4.03. ^aMost common (≥3%) TEAEs leading to dose interruption were anemia/decreased hemoglobin (21.7%), thrombocytopenia/decreased platelets (13.9%), asthenia/fatigue (9.6%), nausea (7.0%), vomiting (6.1%), neutropenia/decreased ANC (6.1%), ALT/AST increased (4.3%), creatinine increased (4.3%), decreased appetite (4.3%), acute kidney injury (3.5%), and hypophosphatemia (3.5%). ^bMost common (≥3%) TEAEs leading to dose reduction were anemia/decreased hemoglobin (13.0%), asthenia/fatigue (9.6%), thrombocytopenia/decreased platelets (7.0%), nausea (6.1%), and decreased appetite (4.3%). ^cOne patient each due to acute respiratory distress syndrome; ALT/AST increased; anemia; balance disorder; cardiac failure; decreased appetite, fatigue, and weight decreased; leukopenia and neutropenia; pneumonia; and prolonged QT. ALT, alanine aminotransferase; ANC, absolute neutrophil count; AST, aspartate aminotransferase; NCI CTCAE, National Cancer Institute-Common Terminology Criteria for Adverse Events; TEAE, treatment-emergent adverse event.	

Supplementary Table S10. Other TEAEs of interest (N=115).

TEAEs, n (%)	Any grade	Grade ≥3
Rash ^{a,b}	31 (27.0)	2 (1.7)
Bleeding ^c	16 (13.9)	4 (3.5)
Cough	15 (13.0)	0
Dyspnea	15 (13.0)	0
Urinary tract infection	15 (13.0)	2 (1.7)
Dysgeusia	14 (12.2)	0
Neutropenia/decreased ANC	12 (10.4)	8 (7.0)
Dyspepsia	9 (7.8)	0
Photosensitivity reaction	8 (7.0)	0
Hypersensitivity ^d	7 (6.1)	0
Sepsis ^e	5 (4.3)	5 (4.3)
Ischemic cardiovascular events ^f	4 (3.5)	3 (2.6)
Leukopenia	4 (3.5)	2 (1.7)
Pneumonia	4 (3.5)	1 (0.9)
Renal failure	4 (3.5)	1 (0.9)
Pulmonary embolism	3 (2.6)	3 (2.6)
Lymphopenia	2 (1.7)	0
Pneumonitis	2 (1.7)	1 (0.9)
Embolism	1 (0.9)	1 (0.9)
Febrile neutropenia	1 (0.9)	1 (0.9)

Visit cutoff date: September 13, 2019.

^aIncludes blister, blood blister, dermatitis, dermatitis contact, eczema, genital rash, palmar-plantar erythrodysesthesia syndrome, photosensitivity reaction, psoriasis, rash, rash maculo-papular, rash pruritic, skin exfoliation, skin lesion, and urticaria.

^bThe combined term of rash led to dose reduction in 4 (3.5%) in patients.

^cIncludes ear hemorrhage, eye hematoma, hematochezia, hematoma, hematuria, hemorrhagic anemia, rectal hemorrhage, and umbilical hemorrhage.

^dIncludes asthma, choking sensation, flushing, periorbital swelling, swelling face, and wheezing.

^eIncludes pulmonary sepsis, sepsis, septic shock, and urosepsis.

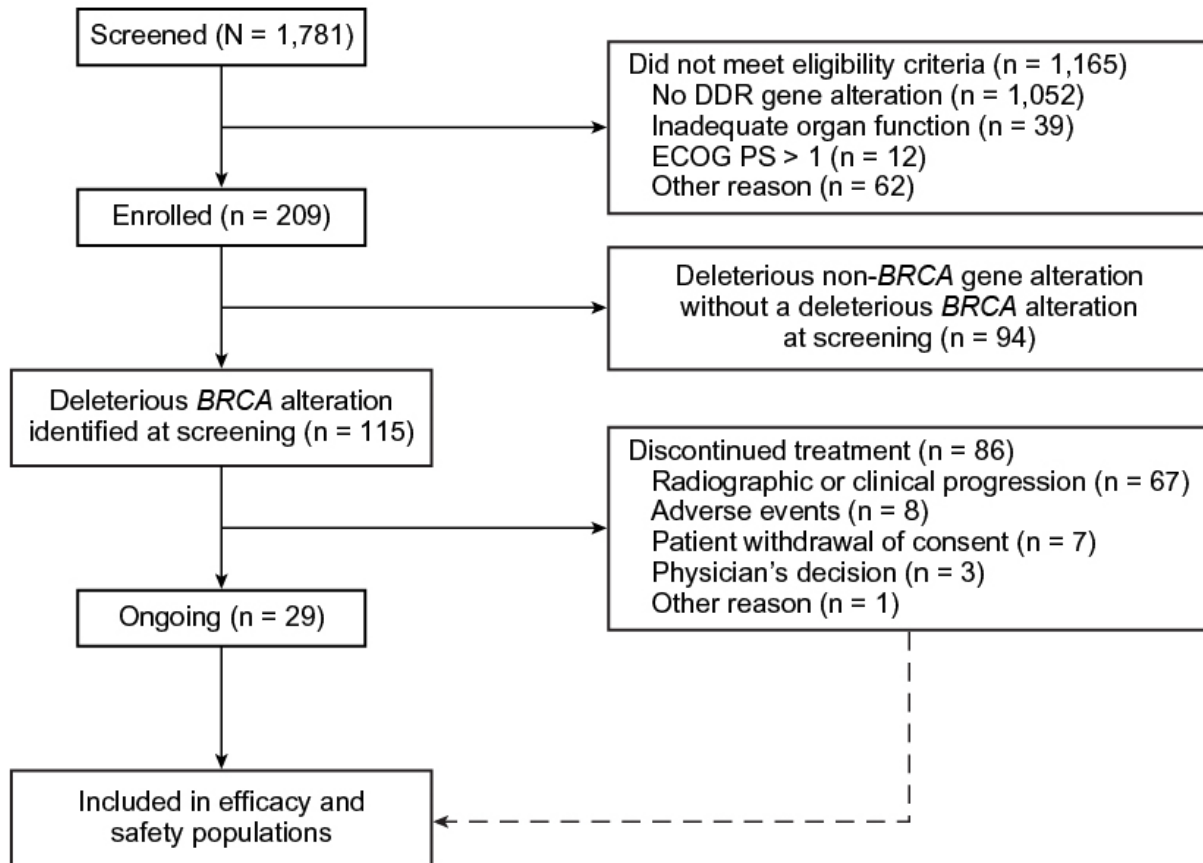
^fIncludes acute coronary syndrome, acute myocardial infarction, angina pectoris, and coronary artery disease.

ANC, absolute neutrophil count; TEAE, treatment-emergent adverse event.

Supplementary Table S11. Laboratory abnormalities reported in $\geq 35\%$ of patients (N=115).

Laboratory parameter, n (%)	Any grade	Grade ≥ 3
Decrease in leukocytes	79 (68.7)	6 (5.2)
Increase in ALT	79 (68.7)	6 (5.2)
Decrease in phosphate	75 (67.6) ^a	17 (15.3) ^a
Decrease in absolute neutrophil count	71 (61.7)	11 (9.6)
Decrease in hemoglobin	68 (59.1)	29 (25.2)
Increase in AST	59 (51.3)	0
Increase in alkaline phosphatase	50 (43.9) ^b	2 (1.8) ^b
Increase in creatinine	49 (42.6)	2 (1.7)
Decrease in lymphocytes	48 (41.7)	19 (16.5)
Increase in triglycerides	41 (41.6) ^c	6 (5.3) ^c
Decrease in platelets	46 (40.0)	11 (9.6)
Decrease in sodium	44 (38.3)	3 (2.6)
Increase in cholesterol	41 (36.3) ^c	1 (0.9) ^c
Visit cutoff date: September 13, 2019. "Any grade" represents any worsening shift from baseline, and "grade 3/4" represents a maximum shift to grade 3/4. NCI CTCAE version 5.0; decrease in phosphate is graded using NCI CTCAE version 4.03 ^aPatients with both baseline and postbaseline results, n=111. ^bPatients with both baseline and postbaseline results, n=114. ^cPatients with both baseline and postbaseline results, n=113. ALT, alanine aminotransferase; AST, aspartate aminotransferase; NCI CTCAE, National Cancer Institute-Common Terminology Criteria for Adverse Events.		

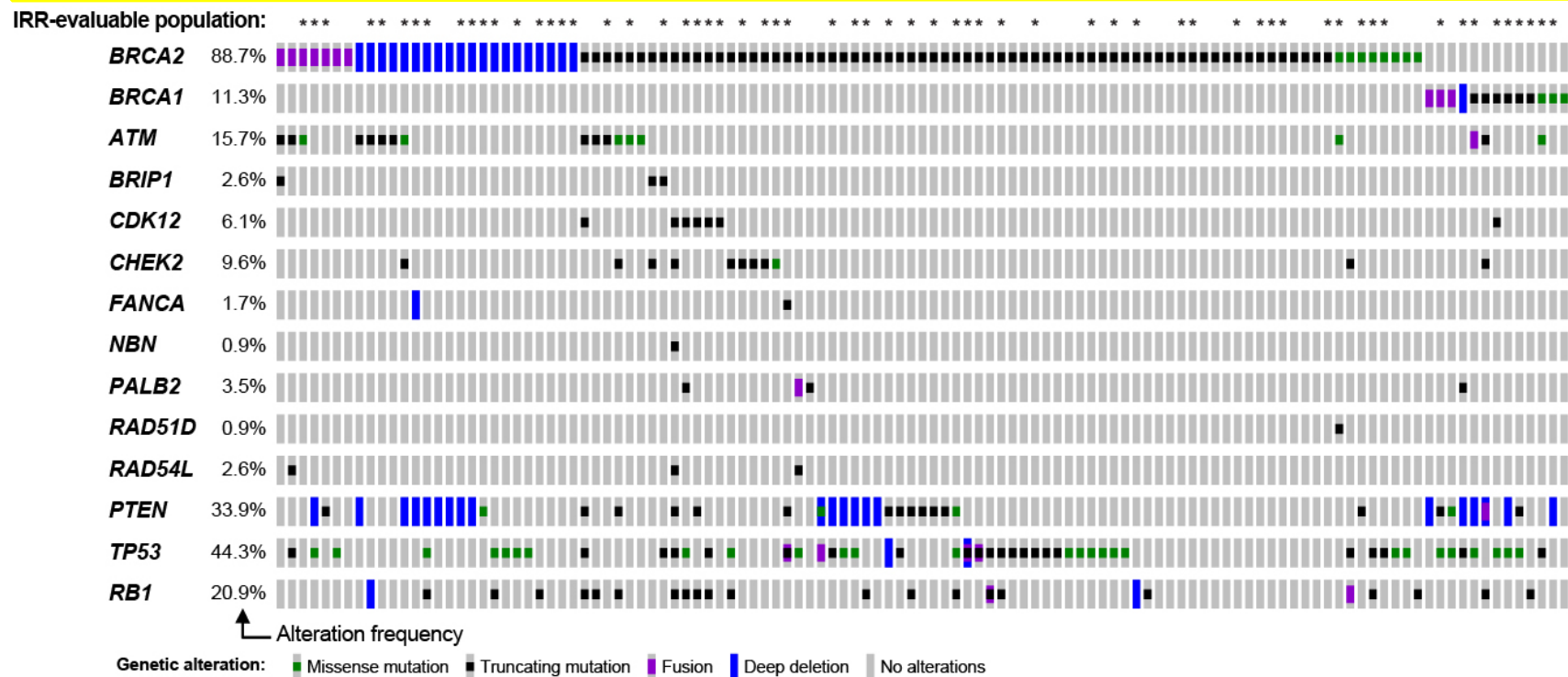
Supplementary Figure S1. CONSORT diagram.



Visit cutoff date: December 23, 2019. Number of patients screened includes patients who entered prescreening and who were still under consideration for study entry as of the May 8, 2019 enrollment cutoff date.

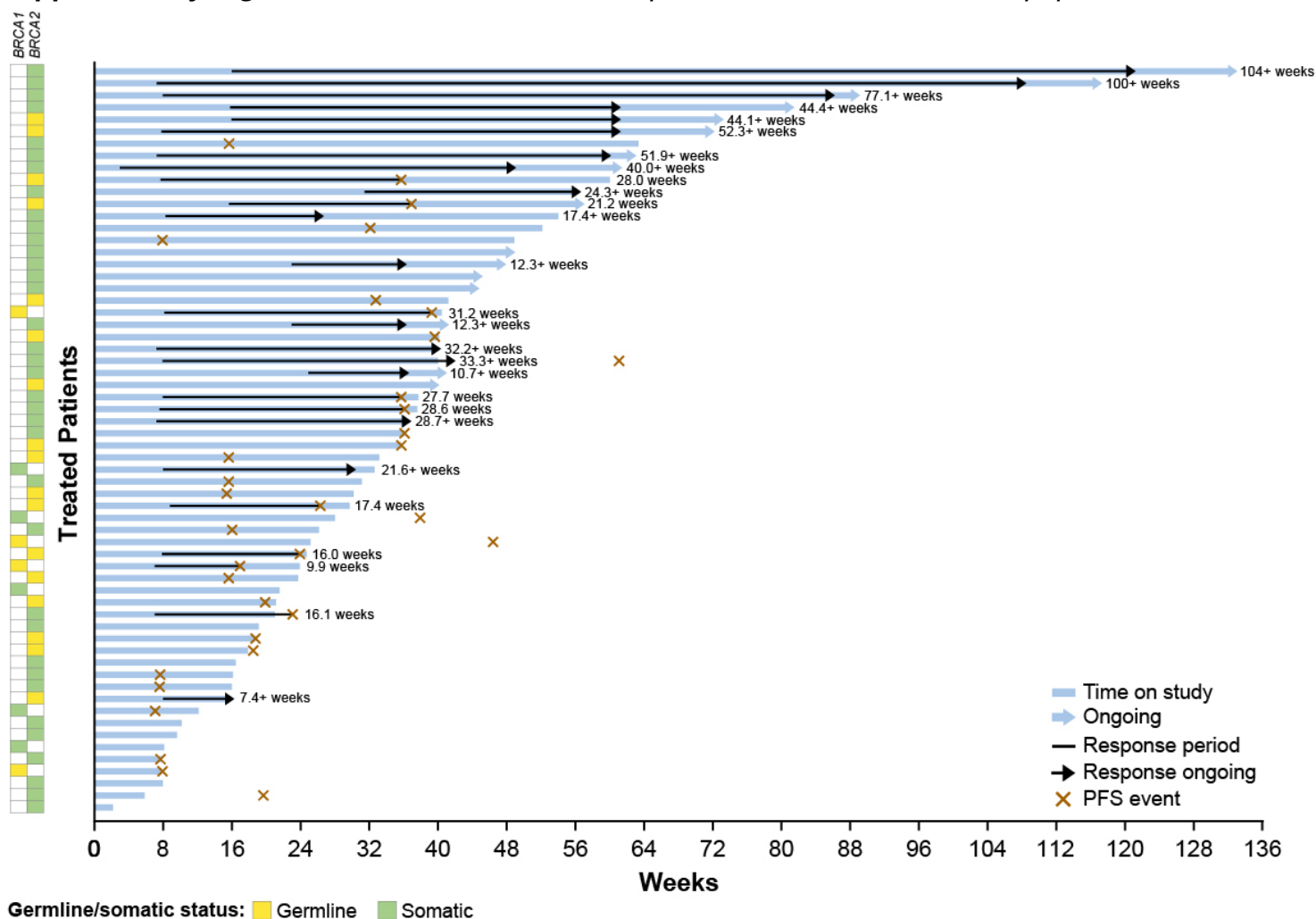
BRCA, *BRCA1* or *BRCA2*; DDR, DNA-damage repair; ECOG PS, Eastern Cooperative Oncology Group Performance Status.

Supplementary Figure S2. Summary of alterations identified among patients in the overall efficacy population (N=115).



IRR, independent radiology review.

Supplementary Figure S3. Treatment duration for patients in the IRR-evaluable population.

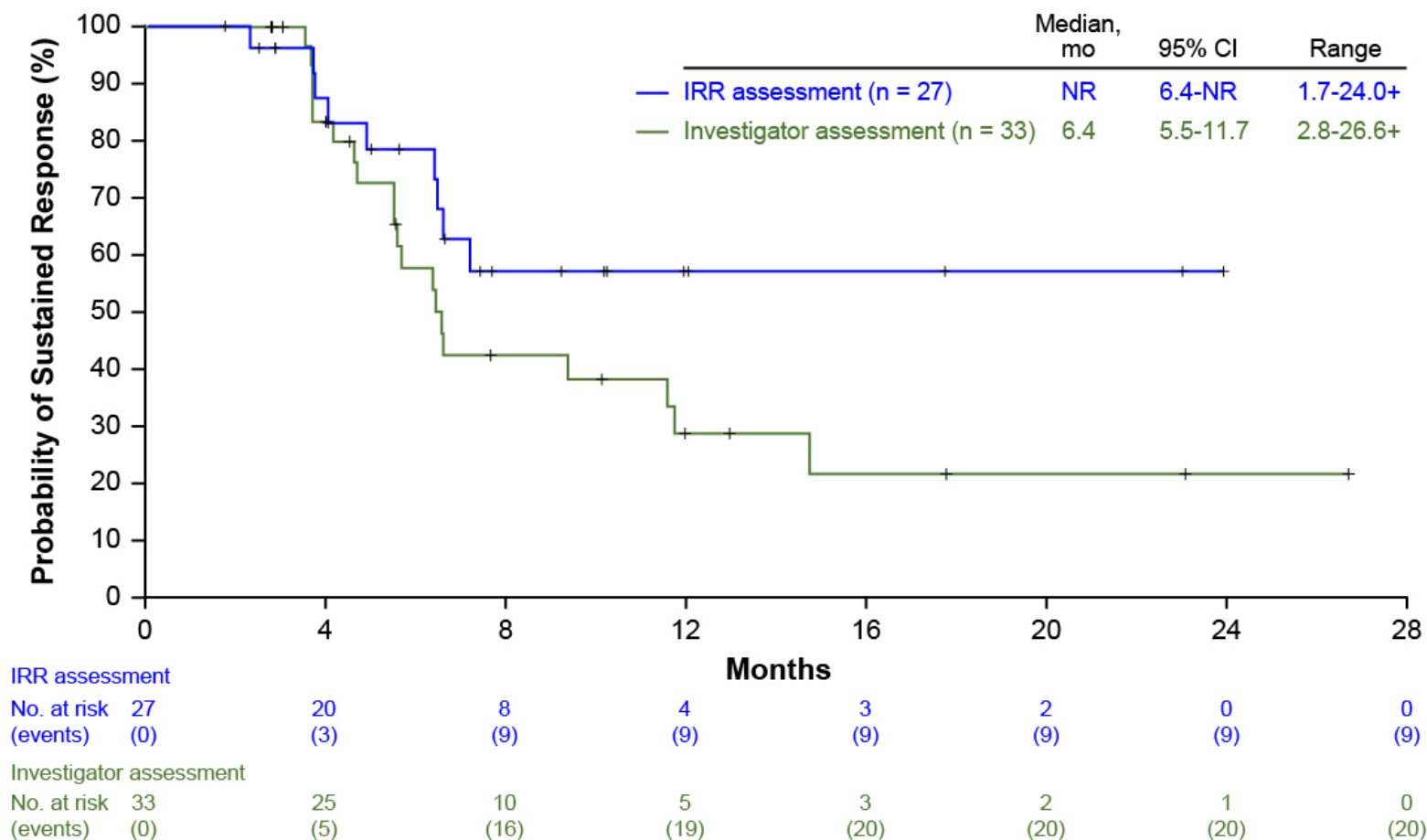


Visit cutoff date: December 23, 2019.

Response was assessed per modified RECIST/PWCG3 criteria. Values in the graph represent DOR.

DOR, duration of response; IRR, independent radiology review; PCWG3, Prostate Cancer Clinical Trials Working Group 3; PFS, progression-free survival; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.

Supplementary Figure S4. DOR for confirmed radiographic response by blinded IRR or investigator in the IRR-evaluable population and investigator-evaluable population, respectively.



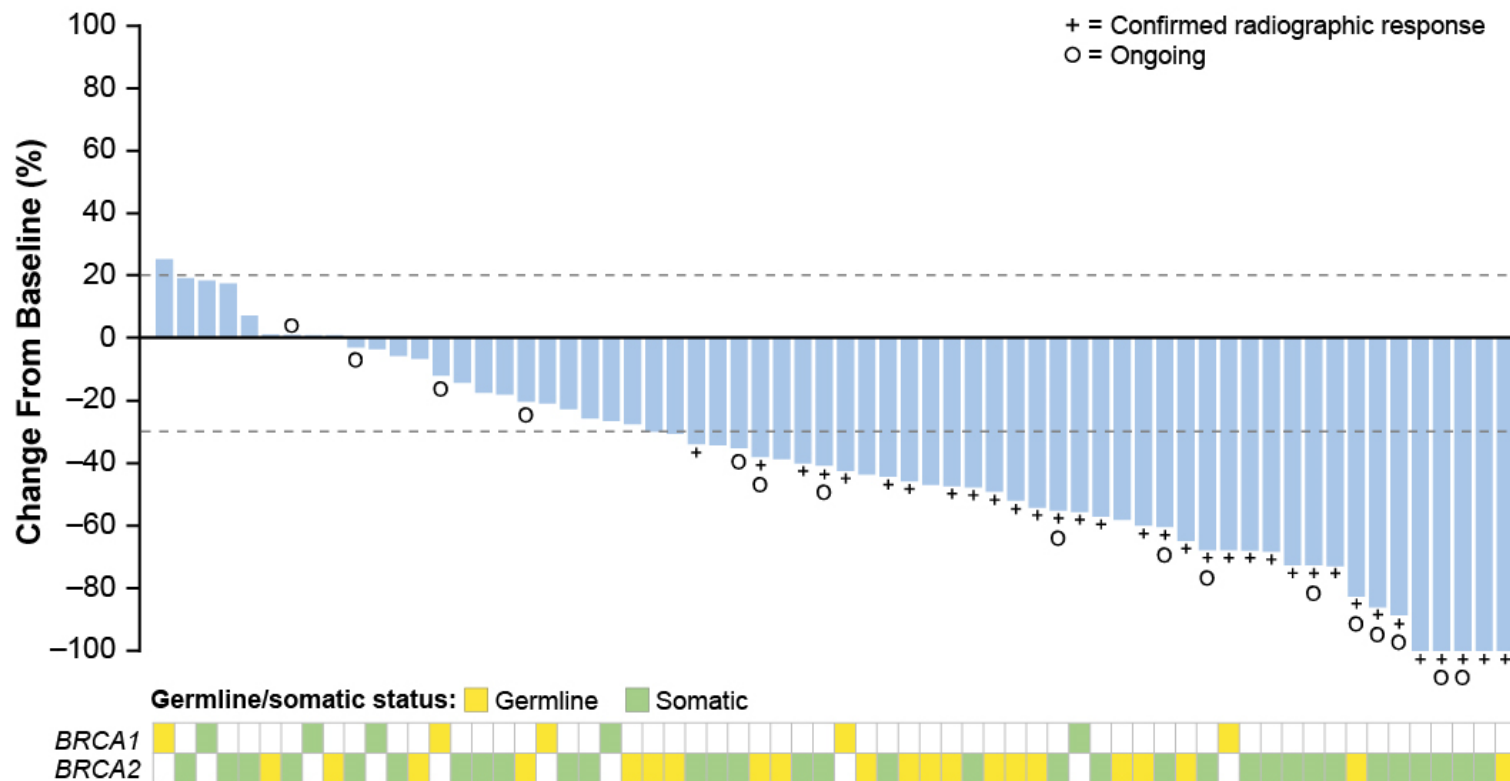
Visit cutoff date: December 23, 2019.

Response was assessed per modified RECIST/PWCG3 criteria.

DOR, duration of response; IRR, independent radiology review; NR, not reached; PCWG3, Prostate Cancer Clinical Trials Working Group 3;

RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.

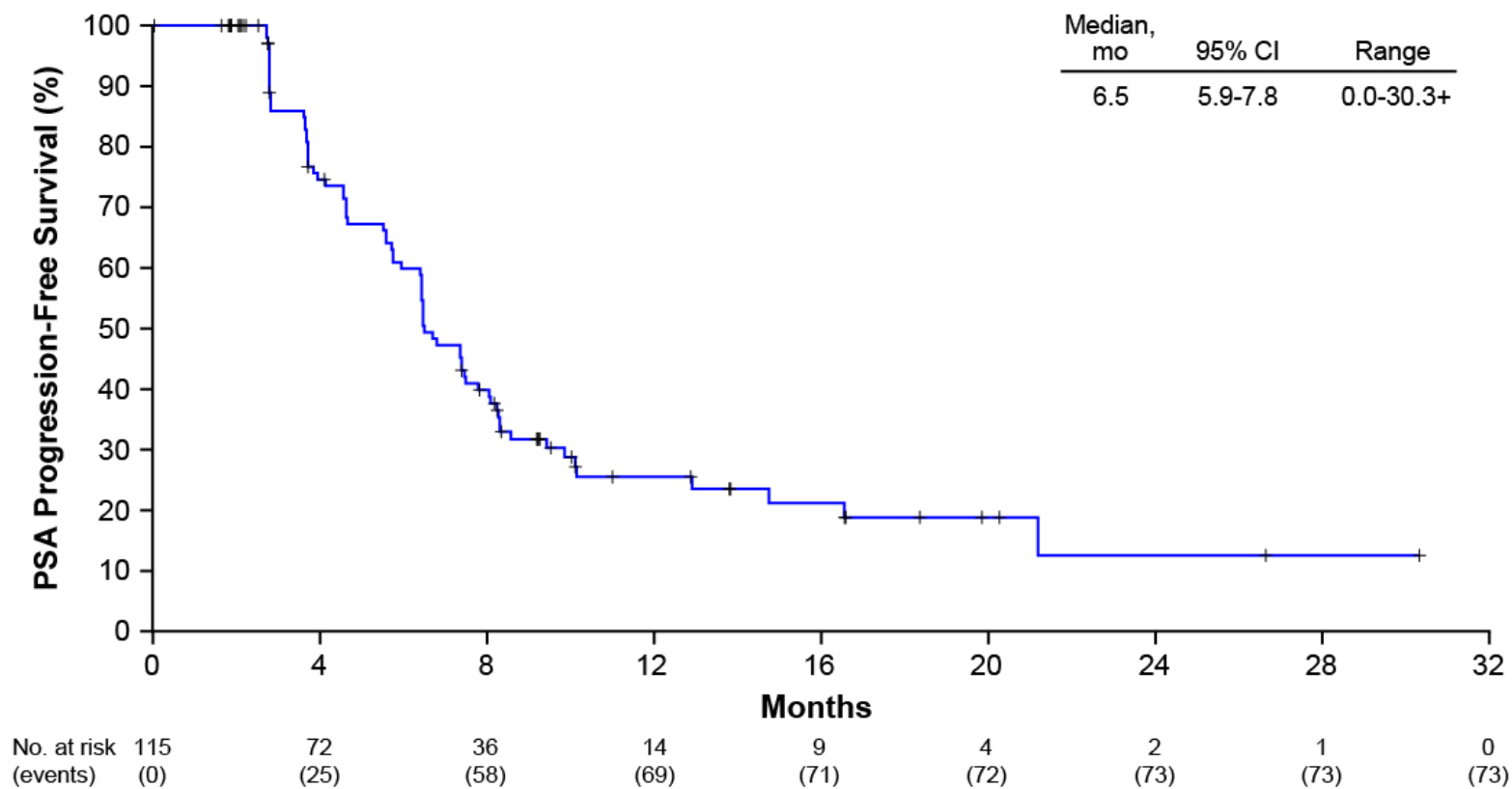
Supplementary Figure S5. Best change from baseline in sum of target lesion(s) in the investigator-evaluable population.



Visit cutoff date: December 23, 2019.

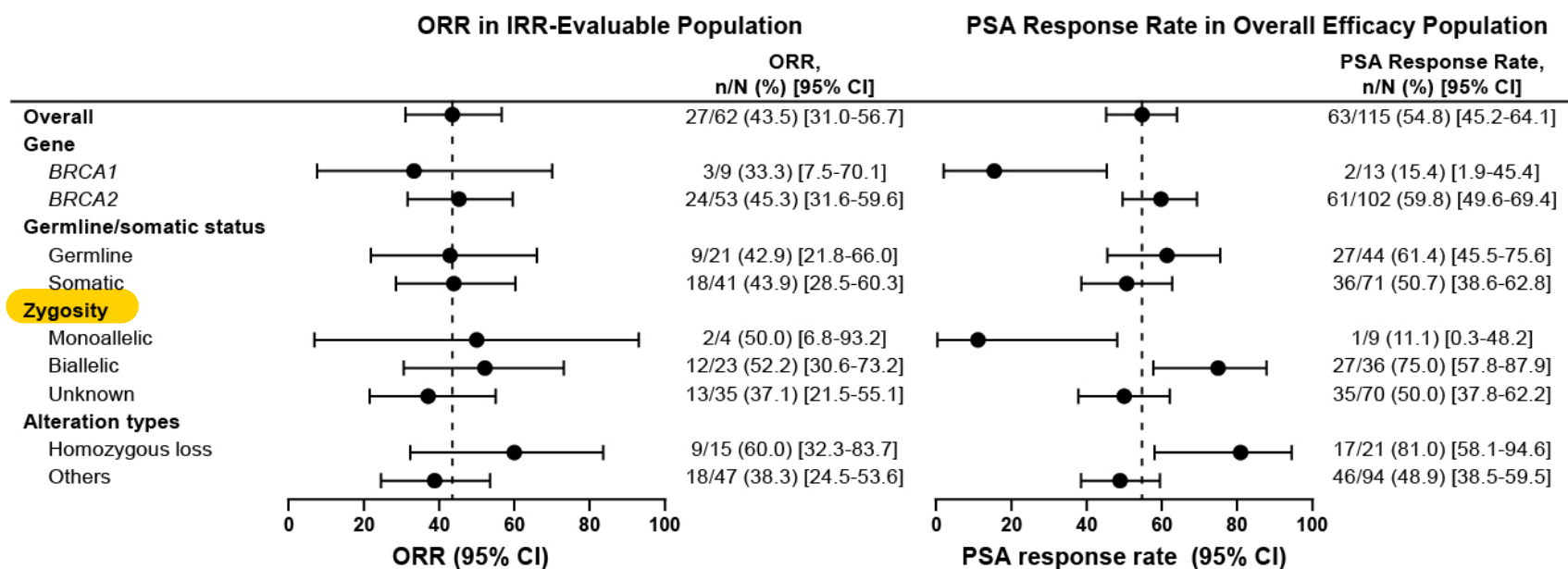
The upper dotted line indicates the threshold for progressive disease, a 20% increase in the sum of the longest diameter of the target lesions, whereas the lower dotted line indicates the threshold for partial response, a 30% decrease in the sum of the longest diameter of the target lesions. Patients with 0% change from baseline are shown as 0.5% for visual clarity.

Supplementary Figure S6. Time to PSA progression in the overall efficacy population.



Visit cutoff date: December 23, 2019.
PSA, prostate-specific antigen.

Supplementary Figure S7. Response by baseline genomic characteristics.

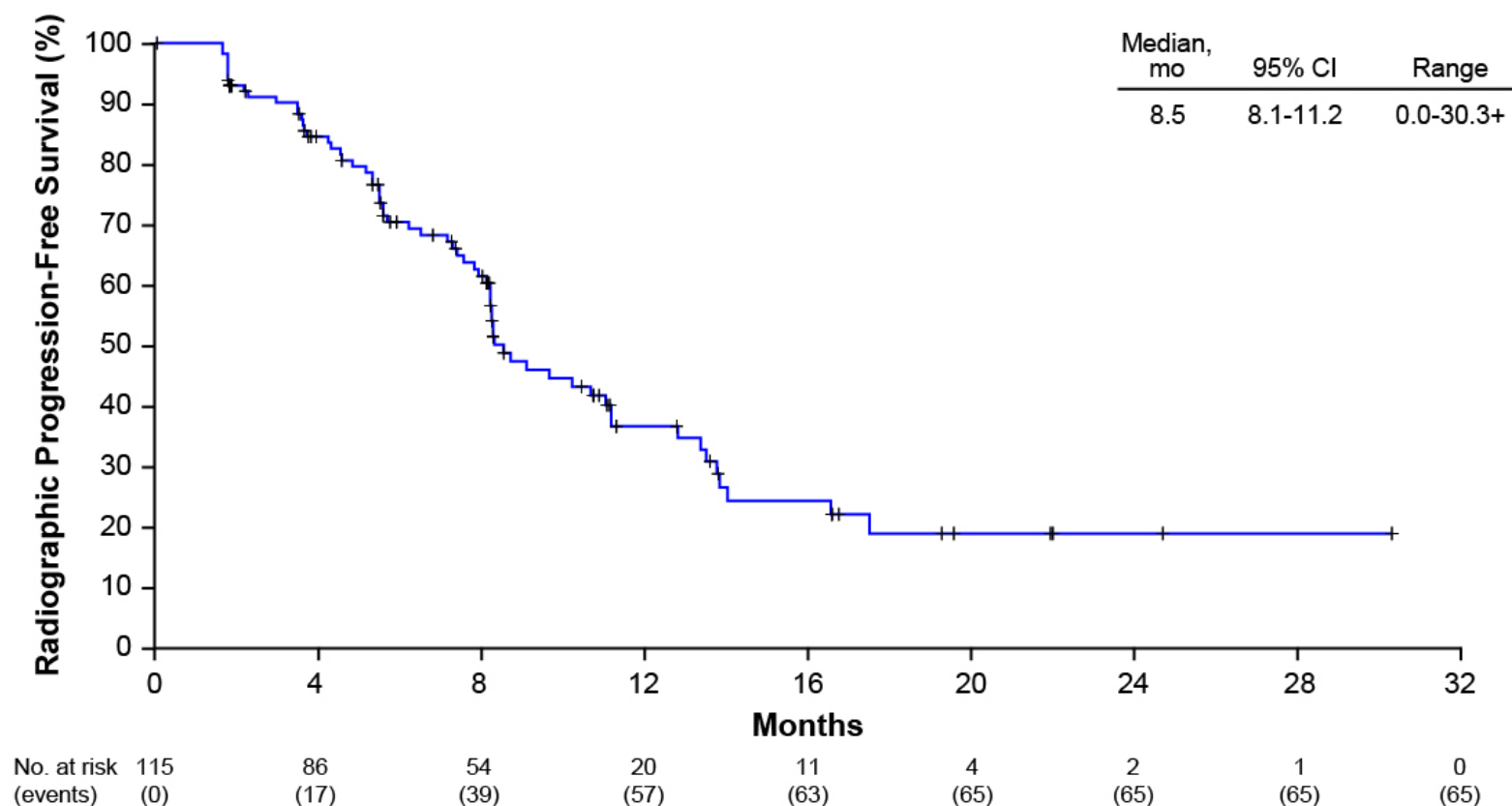


Visit cutoff date: December 23, 2019.

The vertical dotted line corresponds to the overall ORR or PSA response.

IRR, independent radiology review; ORR, objective response rate; PSA, prostate-specific antigen.

Supplementary Figure S8. Radiographic progression-free survival by investigator assessment in the overall efficacy population.

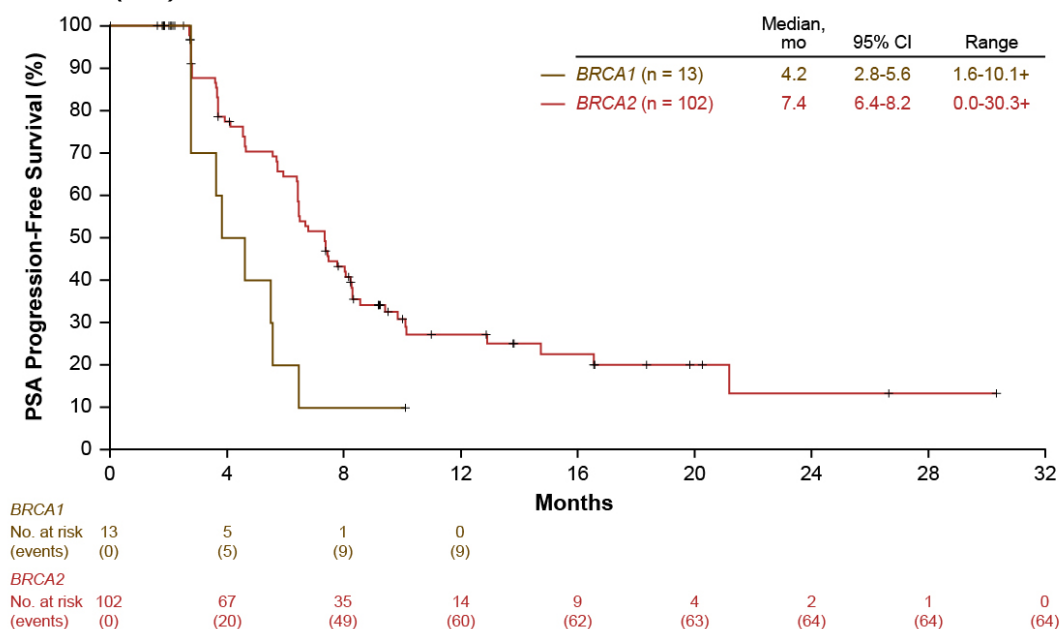


Visit cutoff date: December 23, 2019.

Progression was assessed per modified RECIST/PWCG3 criteria.

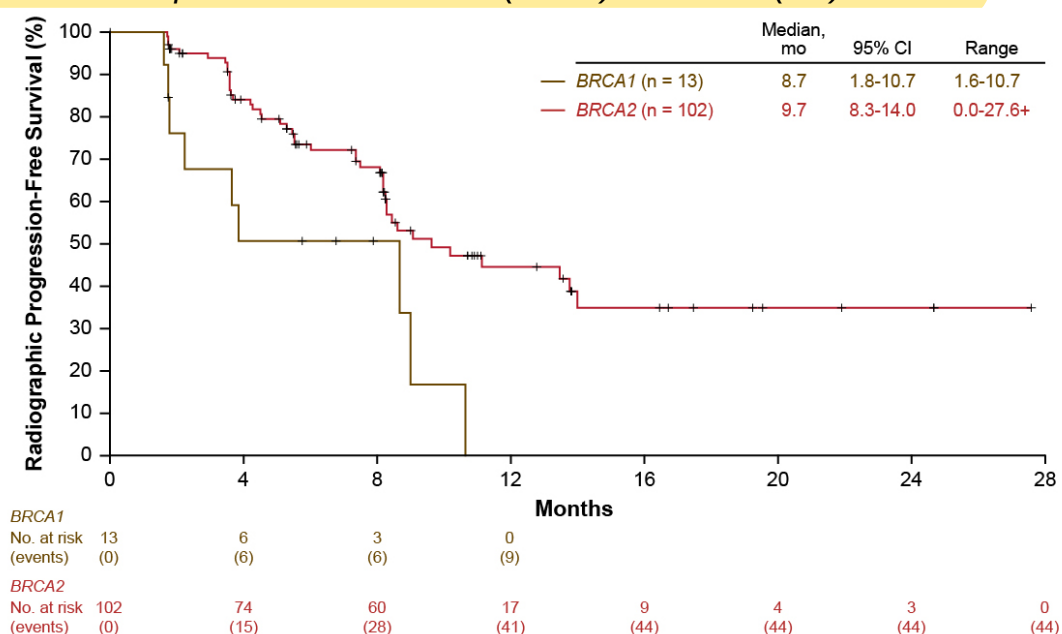
PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.

Supplementary Figure S9. Time to PSA progression in patients with a BRCA1 (brown) or BRCA2 (red) alteration.



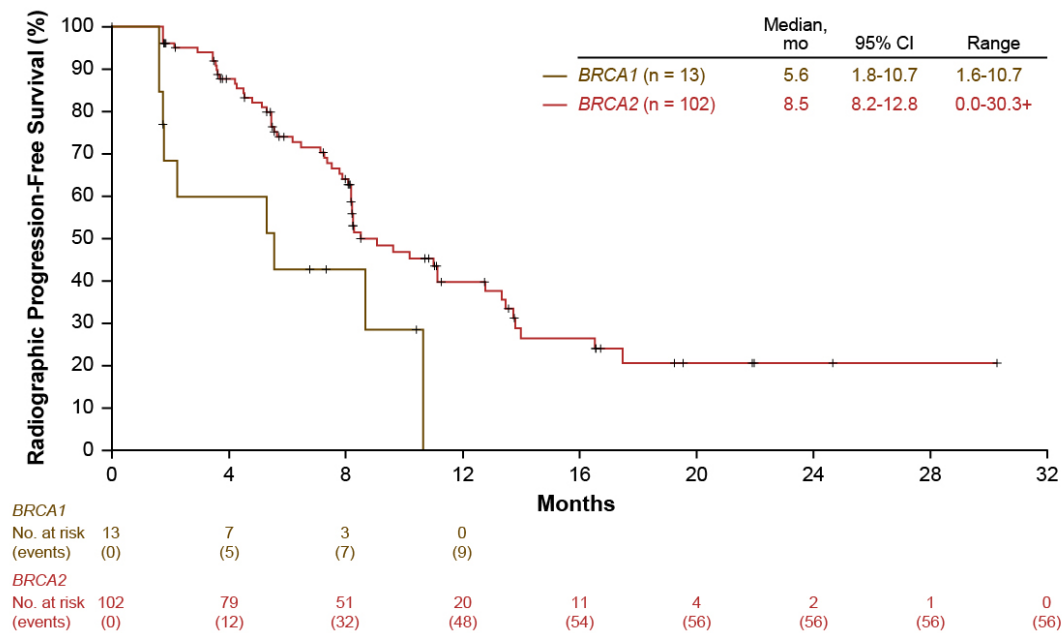
Visit cutoff date: December 23, 2019.
PSA, prostate-specific antigen.

Supplementary Figure S10. Radiographic progression-free survival by blinded IRR assessment in patients with a BRCA1 (brown) or BRCA2 (red) alteration.



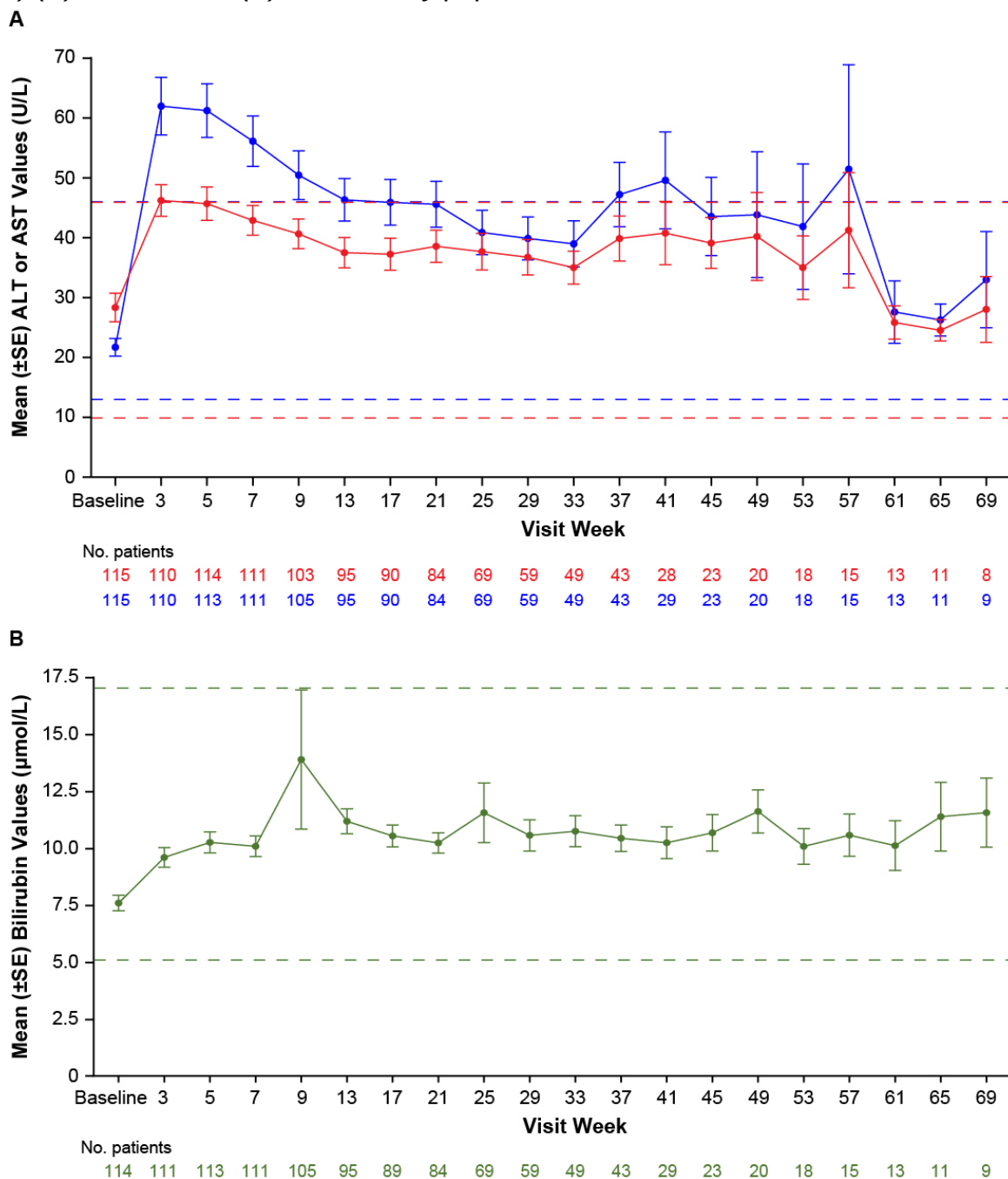
Visit cutoff date: December 23, 2019.
Progression was assessed per modified RECIST/PWCG3 criteria.
IRR, independent radiology review; PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.

Supplementary Figure S11. Radiographic progression-free survival by investigator assessment in patients with a BRCA1 (brown) or BRCA2 (red) alteration.



Visit cutoff date: December 23, 2019.
Progression was assessed per modified RECIST/PWCG3 criteria.
PCWG3, Prostate Cancer Clinical Trials Working Group 3; RECIST, Response Evaluation Criteria In Solid Tumors version 1.1.

Supplementary Figure S12. Baseline and on-treatment value for ALT (blue) and AST (red) (A) and bilirubin (B) in the safety population.

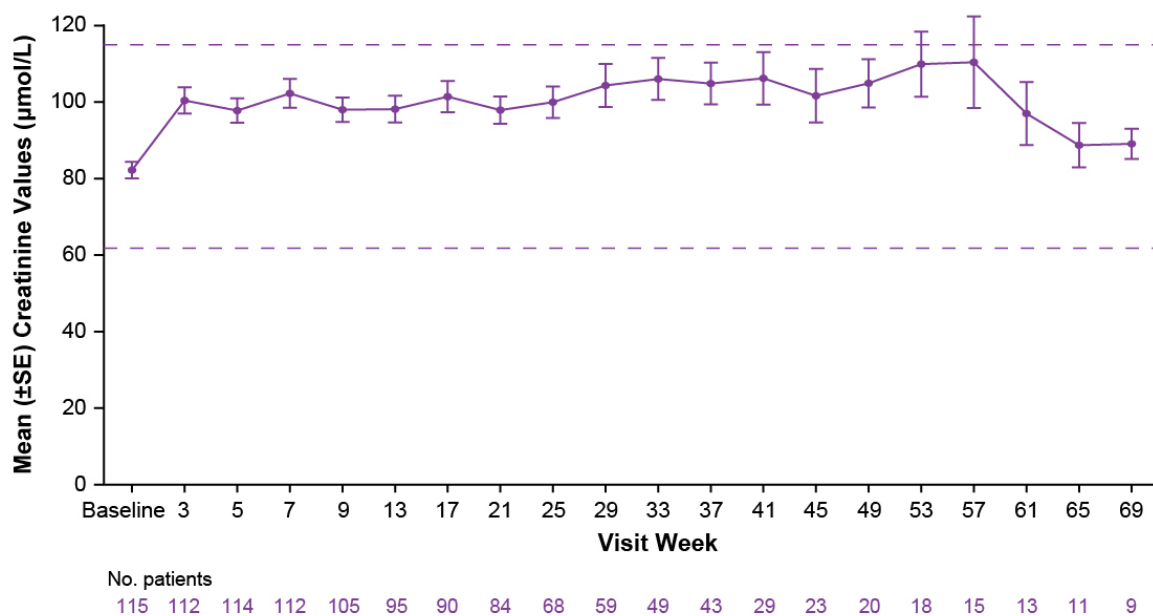


Visit cutoff date: September 13, 2019.

Horizontal lines represent the upper and lower limits of normal for each laboratory parameter. Error bars represent standard error of the mean.

ALT, alanine aminotransferase; AST, aspartate aminotransferase; SE, standard error.

Supplementary Figure S13. Baseline and on-treatment value for creatinine in the safety population.



Visit cutoff date: September 13, 2019.

Horizontal lines represent the upper and lower limits of normal. Error bars represent standard error of the mean.

SE, standard error.